Supplementary Material: An Explainable Hybrid of Lexicon Rules and Multilingual Transformers for Bilingual Hate Speech Detection

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1 Overview of Supplementary Experiments

This document provides supplementary materials for the ICCIES 2027 paper (15-page main manuscript). It details exhaustive per-component experimental evaluations, including language routing, lexicon rule ablation, neural backbone comparisons, decision-level fusion, end-to-end pipeline benchmarks, latency breakdowns, surface form robustness, error overlap, and paired statistical significance testing (bootstrap and McNemar tests). All reported empirical figures are extracted directly from frozen evaluation outputs stored in `outputs/evaluation/`.

2 Language Detection and Routing

2.1 Language Routing Confusion Matrix

Table 1 presents the confusion matrix for the language router evaluated across the 9,158-sample bilingual test stream (6,680 ViHSD comments + 2,478 Davidson comments).

Table 1. Language detection routing confusion matrix (n = 9,158). Rows: Actual ground-truth language; Columns: Router output decision.

Actual Language	Predicted VI	Predicted EN	Predicted Mix	Predicted Multi.	Total
Vietnamese	6,278	58	27	317	6,680
English	4	2,341	4	129	2,478
Total	**6,282**	**2,399**	**31**	**446**	**9,158**

2.2 Language Detection Accuracy on Clean Held-out Sets

Table 2 reports routing accuracy on the clean ViNLI (Vietnamese) and Wortschatz (English) held-out sets used for the language detection benchmark.

Table 2. Language detection accuracy on clean held-out sets (ViNLI Vietnamese, $n = 29{,}934$; Wortschatz English Leipzig 2015, $n = 29{,}934$).

Test Set	N	Detection Accuracy (%)
ViNLI (Vietnamese)	29,934	99.97%
Wortschatz (English)	29,934	99.64%
Overall	**59,868**	**99.81%**

2.3 Routing Latency Breakdown

Table 3 details the latency profile for the language routing stage. The Language Router row is measured on the combined ViNLI + Wortschatz clean benchmark ($n = 59{,}868$); the AI Backbone and Total rows retain the original 9,158-stream measurements.

Table 3. Routing latency breakdown (Language Router row on ViNLI+Wortschatz clean benchmark, $n = 59{,}868$).

Stage	Mean (ms)	P50 (ms)	P95 (ms)	P99 (ms)	Throughput (texts/s)
Language Router (langID)	1.668	1.633	1.898	2.165	599.5
AI Backbone Pipeline	3.990	—	—	—	250.6
Total Serialised Pipeline	**5.658**	—	—	—	—

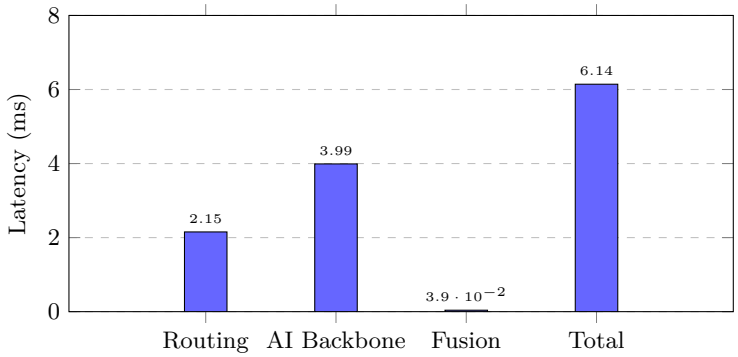


Fig. 1. Latency breakdown by pipeline stage on the bilingual test stream ($n = 9{,}158$).

3 Rule Engine Component Ablation Study

3.1 Component Contribution Evaluation

Table 4 reports component ablation metrics for the Vietnamese Lexicon Rule Engine evaluated on the ViHSD test split ($n = 6{,}680$). Each variant disables one rule mechanism, re-tunes thresholds on validation, and scores test once.

Table 4. Vietnamese Lexicon Rule Engine component ablation study on ViHSD test ($n = 6{,}680$).

Variant	Accuracy	Macro-F1	HATE F1	ΔMacro-F1 vs Full
Full Rule Engine	0.8774	0.6602	0.5906	—
No Entry Confidence	0.8781	0.6610	0.5935	+0.0008
No Context Matching	0.8774	0.6602	0.5906	0.0000
No Obfuscation Views	0.8772	0.6590	0.5896	-0.0012
No Severity Scaling	0.8774	0.6602	0.5906	0.0000

3.2 Detailed Rule Engine Confusion Matrices

Table 5 illustrates misclassification patterns across rule engine variants on ViHSD test.

Table 5. Detailed confusion matrix breakdown for standalone Rule Engine variants on ViHSD test ($n = 6{,}680$).

Rule Variant	HATE → HATE	HATE → OFF	HATE → CLEAN	OFF → CLEAN
Full (argmax)	348	0	340	272
No Fuzzy Matching	348	0	340	272
No Diacritics Match	210	160	318	259
No Normalisation	0	0	688	444
Full + Frozen Thresholds	210	160	318	259

3.3 Per-Class Metrics for Rule Engine

Table 6 presents class-wise precision, recall, and F1 scores for rule engine configurations.

Table 6. Per-class precision, recall, and F1 metrics for standalone Rule Engine configurations on ViHSD test.

Configuration	HATE Precision	HATE Recall	HATE F1	OFF F1	CLEAN F1
Full Engine (argmax)	0.469	0.506	0.487	0.000	0.927
Full Engine (thresholds)	0.625	0.305	0.410	0.239	0.928

4 Neural Backbone Model Analysis

4.1 Vietnamese Backbone Models

Table 7 summarizes standalone test performance across six Vietnamese Transformer backbones on ViHSD test (n = 6,680).

Table 7. Performance comparison of Vietnamese mDeBERTa-v3 backbones on ViHSD test (n = 6,680).

Model Identifier	Accuracy	Macro-F1	HATE F1	OFF F1	ECE
M1 (`sensitiveai-vi`)	0.8734	0.6545	0.5864	0.4367	0.106
M2 (`sensitiveai-vi-custom`)	0.8762	0.6575	0.5830	0.4493	0.111
M3 (`sensitiveai-vi-customdlr2stage`)	0.8754	0.6638	0.5943	0.4561	0.120
M4 (`sensitiveai-vi-custom-curriculum`)	0.8731	0.6590	0.5847	0.4535	0.119
M5 (`sensitiveai-vi-customdlr2stage-ext`)	0.8783	0.6536	0.5843	0.4350	0.118
M6 (`sensitiveai-vi-stage2`)	0.8633	0.6604	0.6088	0.4382	0.068

4.2 English Backbone Models

Table 8 evaluates English Transformer backbones fine-tuned on the Davidson test split (n = 2,478).

Table 8. Performance comparison of English backbones on Davidson test (n = 2,478).

Model Identifier	Accuracy	Macro-F1	HATE F1	OFF F1	CLEAN F1
`sensitiveai-en`	0.9031	0.7676	0.4554	0.9410	0.9063
`sensitiveai-en-custom`	0.9031	0.7587	0.4377	0.9404	0.8980
`sensitiveai-en-custom-curriculum`	0.9072	0.7441	0.3908	0.9445	0.8970
`sensitiveai-en-customdlr2stage`	0.9108	0.7509	0.4048	0.9457	0.9024
`sensitiveai-en-stage2`	0.9080	0.7722	0.4656	0.9440	0.9070

4.3 ROC-AUC and Average Precision Performance

Table 9 details multi-class ROC-AUC metrics across representative backbone checkpoints.

Table 9. Class-wise ROC-AUC scores evaluated on ViHSD test split (n = 6,680).

Model Checkpoint	HATE ROC-AUC	OFF ROC-AUC	CLEAN ROC-AUC	Macro ROC-AUC
`sensitiveai-vi-custom-curriculum`	0.8949	0.8638	0.8924	0.8837
`sensitiveai-vi-customdlr2stage`	0.8966	0.8635	0.8924	0.8842
`sensitiveai-vi-custom`	0.8951	0.8640	0.8920	0.8837

5 Decision-Level Fusion Variants

5.1 Comprehensive Decision Strategy Comparison

Table 10 compares all evaluated decision-level fusion strategies on ViHSD test using the frozen `sensitiveai-vi-custom` backbone.

Table 10. Comprehensive evaluation of decision fusion rules on ViHSD test (n = 6,680).

Decision Strategy	Accuracy	Macro-F1	HATE F1	OFF F1	CLEAN F1
AI-only (argmax)	0.8762	0.6575	0.5830	0.4493	0.9403
Rule-only (argmax)	0.8494	0.4714	0.4867	0.0000	0.9274
Rule-only (frozen thresholds)	0.8421	0.5258	0.4102	0.2389	0.9284
Fusion, global α (argmax)	0.8769	0.6583	0.5851	0.4490	0.9408
Fusion, global α (thresholds)	0.8769	0.6583	0.5851	0.4490	0.9408
Fusion, logistic meta-classifier	0.8624	0.6511	0.5815	0.4379	0.9340
Fusion, per-channel α (argmax)	0.8774	0.6593	0.5882	0.4488	0.9408
Fusion, per-channel α + thresholds	0.8774	**0.6602**	0.5906	0.4494	0.9407

5.2 Calibration Diagnostics (ECE, Brier, NLL)

Table 11 summarizes probability calibration metrics across system configurations.

Table 11. Calibration diagnostics (ECE, Brier score, Negative Log-Likelihood, mean confidence) across evaluation splits.

System Variant	ECE ↓	Brier ↓	NLL ↓	Mean Conf.	Macro-F1 ↑
AI-only (ViHSD test)	0.111	0.2334	0.8382	0.987	0.6575
Hybrid Fusion (ViHSD test)	0.098	0.2273	0.7879	0.952	0.659
Uniform Control ($\alpha = 0.70$)	0.092	0.227	—	0.791	0.658
Mixed E2E Stream (n = 9,158)	0.078	0.230	0.755	0.904	0.779
AI-only M3 (n = 6,680)	0.106	0.228	0.622	0.979	0.655
Hybrid M3 (n = 6,680)	0.068	0.217	0.482	0.931	0.660

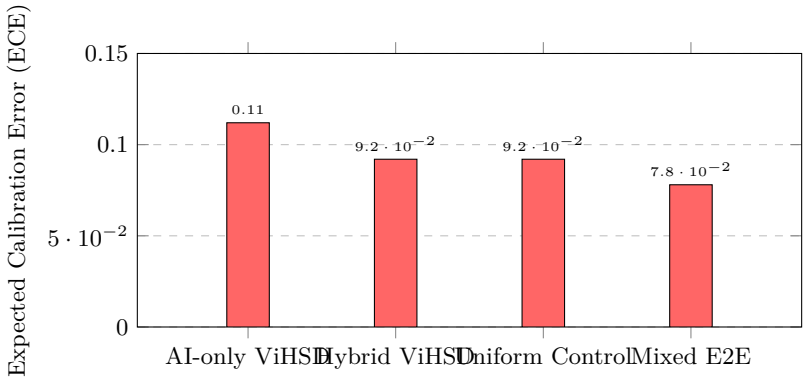


Fig. 2. Expected Calibration Error (ECE) comparison between AI-only, Hybrid, and baseline control variants.

6 End-to-End Pipeline Performance

6.1 Full End-to-End Benchmark Results

Table 12 provides complete end-to-end performance metrics over the 9,158 bilingual test stream.

Table 12. Full end-to-end pipeline benchmark on the bilingual test split ($n = 9{,}158$).

Execution Mode	Stream / Language	N	Accuracy	Macro-F1	ECE
Backbone-only	Vietnamese (ViHSD)	6,680	0.8713	0.6602	0.112
Backbone-only	English (Davidson)	2,478	0.8717	0.7273	0.110
AI argmax E2E	Bilingual Stream	9,158	0.8714	0.7792	0.101
AI thresholds E2E	Bilingual Stream	9,158	0.8719	0.7803	0.101
Hybrid E2E Pipeline	Bilingual Stream	**9,158**	**0.8728**	**0.7787**	**0.078**
AI argmax Oracle	Bilingual Stream	9,158	0.8812	0.7898	0.092
AI thresholds Oracle	Bilingual Stream	9,158	0.8817	0.7910	0.093
Hybrid Oracle	Bilingual Stream	9,158	0.8826	0.7896	0.070

6.2 Per-Language Breakdown of Hybrid E2E

Table 13 breaks down Hybrid E2E results per constituent language split.

Table 13. Language-stratified metrics for Hybrid E2E pipeline vs Oracle routing baselines.

Evaluation Split	N	Accuracy	Macro-F1	HATE F1	OFF F1	CLEAN F1
Vietnamese (Hybrid E2E)	6,680	0.8732	0.6603	0.6098	0.4322	0.9388
English (Hybrid E2E)	2,478	0.8717	0.7313	0.4525	0.9196	0.8218
Vietnamese (AI Oracle)	6,680	0.8711	0.6608	0.6197	0.4251	0.9375
English (AI Oracle)	2,478	0.9084	0.7676	0.4505	0.9439	0.9083
Vietnamese (Hybrid Oracle)	6,680	0.8731	0.6609	0.6119	0.4320	0.9387
English (Hybrid Oracle)	2,478	0.9084	0.7726	0.4656	0.9443	0.9081

6.3 Full Stream Per-Label Performance

Table 14 presents per-class precision, recall, and F1 across decision modes for the entire stream.

Table 14. Class-wise performance metrics on the full 9,158-sample bilingual stream.

Pipeline Mode	Target Class	Precision	Recall	F1-Score	Support (N)
AI argmax E2E	HATE	0.5931	0.5788	0.5859	831
	OFFENSIVE	0.8502	0.7981	0.8233	2,362
	CLEAN	0.9158	0.9412	0.9283	5,965
AI thresholds E2E	HATE	0.5889	0.5860	0.5875	831
	OFFENSIVE	0.8561	0.7959	0.8249	2,362
	CLEAN	0.9157	0.9418	0.9286	5,965
Hybrid E2E Pipeline	HATE	0.5920	0.5692	0.5804	831
	OFFENSIVE	0.8592	0.7959	0.8264	2,362
	CLEAN	0.9140	0.9455	0.9295	5,965

6.4 Full Stream Normalized Confusion Matrix

Table 15 details row-normalized recall confusion for the Hybrid E2E pipeline.

Table 15. Normalized confusion matrix (row recall) for Hybrid E2E on 9,158 samples. Rows: Actual; Cols: Predicted.

Actual → Predicted	HATE	OFFENSIVE	CLEAN	Total	HATE Recall	OFF Recall	CLEAN Recall
HATE	473	135	223	831	0.569	0.162	0.268
OFFENSIVE	174	1,880	308	2,362	0.074	0.796	0.130
CLEAN	152	150	5,640	5,965	0.025	0.025	0.946

7 Statistical Significance Testing

7.1 Paired Bootstrap and McNemar Tests

Table 16 presents statistical significance testing results across model pairs (10,000 bootstrap resamples, seed 42).

Table 16. Paired statistical significance tests (paired bootstrap and exact McNemar test) on ViHSD test ($n = 6{,}680$).

Model Comparison (A - B)	ΔMacro-F1	p-F1 (Bootstrap)	ΔAccuracy	p-value (McNemar)
M3 vs M1	+0.0092	0.298	+0.0021	0.526
M3 vs M2	+0.0062	0.328	-0.0007	0.788
M3 vs M4	+0.0047	0.342	+0.0024	0.217
M3 vs M6	+0.0033	0.678	+0.0121	<0.001
M3 vs M5	+0.0101	0.113	-0.0028	0.217
M5 vs M1	-0.0009	0.906	+0.0049	0.110
M6 vs M1	+0.0059	0.377	-0.0100	<0.001
Hybrid E2E vs AI argmax E2E	-0.0004	0.841	+0.0014	0.165

7.2 Model Error Complementarity and Overlap

Table 17 reports the error overlap distribution across all six Vietnamese neural backbones on ViHSD test.

Table 17. Distribution of misclassified comment counts across all six Vietnamese neural backbones ($n = 6{,}680$).

Number of Incorrect Backbones	0	1	2	3	4	5	6
Comment Count	5,410	228	178	93	143	124	504

7.3 Prior-Reweighting Oracle Diagnostics

Table 18 presents class prior reweighting oracle gains on ViHSD test splits.

Table 18. Prior-reweighting oracle diagnostic across Vietnamese backbones on ViHSD test.

Model Checkpoint	Standard Argmax M-F1	Optimal Weight (u^*)	Oracle M-F1	Gain (Δ)
M1 (`sensitiveai-vi`)	0.6545	0.70	0.6564	+0.0018
M2 (`sensitiveai-vi-custom`)	0.6575	0.80	0.6585	+0.0009
M3 (`sensitiveai-vi-customdlr2stage`)	0.6638	1.00	0.6638	+0.0000
M4 (`sensitiveai-vi-custom-curriculum`)	0.6602	2.80	0.6616	+0.0014
M5 (`sensitiveai-vi-customdlr2stage-ext`)	0.6536	3.00	0.6549	+0.0012
M6 (`sensitiveai-vi-stage2`)	0.6604	0.60	0.6656	+0.0051

8 Surface Form and Error Analysis

8.1 Surface Form Robustness Stratification

Table 19 details performance stratification across challenging noise and orthographic surface phenomena on ViHSD test.

Table 19. Performance breakdown across text surface categories on ViHSD test split ($n = 6{,}680$). R: Recall.

Surface Form Category	N	Accuracy	Macro-F1	HATE R	OFF R	CLEAN R
Teencode / Slang	1,039	0.849	0.630	0.453	0.436	0.942
Unaccented / Drop Diacritics	370	0.849	0.634	0.532	0.346	0.943
Repeated Chars (Leet-like)	997	0.882	0.699	0.625	0.561	0.935
Emoji Present	974	0.912	0.616	0.426	0.367	0.972
Very Short Text (< 20 chars)	1,852	0.924	0.709	0.553	0.534	0.972
Very Long Text (> 500 chars)	145	0.759	0.568	0.741	0.143	0.821
Standard Accent Diacritics	6,114	0.876	0.656	0.532	0.418	0.955
Total Test Split	**6,680**	**0.871**	**0.661**	—	—	—

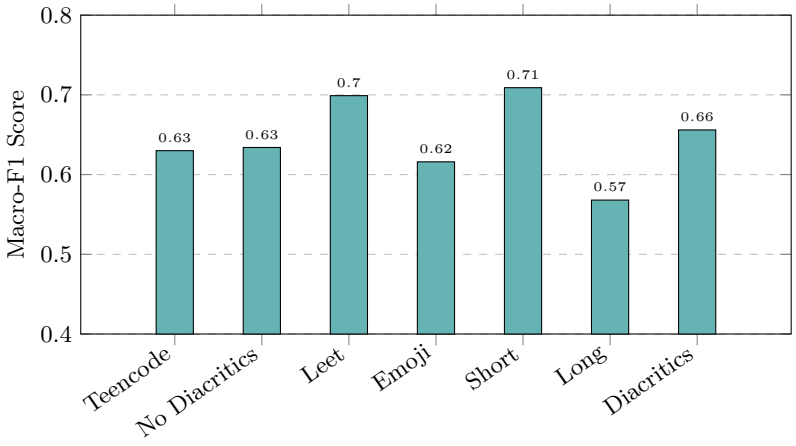


Fig. 3. Macro-F1 performance comparison across noise and surface form categories.

8.2 Analysis of the 504 Blind-Spot Samples

Table 20 analyzes ground-truth label distributions for the 504 test comments missed simultaneously by all six neural backbones.

Table 20. Class distribution of the 504 blind-spot test comments missed by all six neural backbones on ViHSD.

Ground-Truth Class	Sample Count (N)	Percentage (%)
HATE	185	36.7%
OFFENSIVE	180	35.7%
CLEAN	139	27.6%
Total Blind-Spot	**504**	**100.0%**

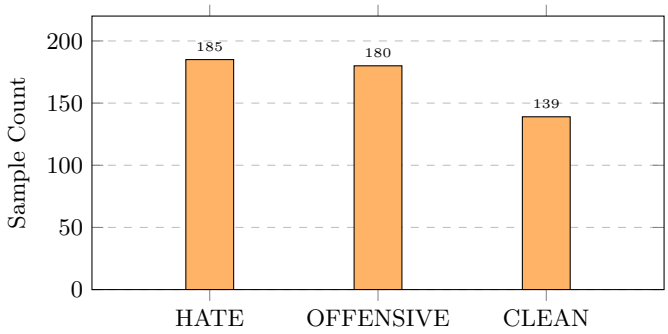


Fig. 4. Class distribution within the 504 blind-spot comments missed by all six neural backbones.

8.3 Decision Change Analysis

Table 21 summarizes decision modifications introduced by Hybrid Fusion relative to AI-only argmax.

Table 21. Summary of decision changes introduced by Hybrid Fusion relative to AI-only argmax ($n = 9{,}158$).

Metric / Category	Value (N)	Rate (%)
Total Changed Decisions	91	0.99%
Corrections (Wrong → Correct)	44	48.4%
Regressions (Correct → Wrong)	31	34.1%
Net Gain	+13	+0.14%
ViHSD Changed Decisions ($n = 6{,}680$)	64	0.96%

9 Experimental and Hardware Environment

Table 22 outlines exact computational infrastructure, software versions, and training protocols.

Table 22. Summary of experimental hardware, software dependencies, and training protocol configuration.

Category	Configuration Details
Hardware	NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM), Intel Core i7-13650HX, 24 GB RAM
Operating System	Windows 11 Home 64-bit
Software Stack	Python 3.11.9, PyTorch 2.5.1, Hugging Face Transformers 4.46.3, scikit-learn 1.5.2
Backbone Architecture	`microsoft/mdeberta-v3-base` (12-layer, 768-hidden, 12-heads, 278M parameters)
Precision & Batch	Single-precision FP32, Evaluation Batch Size = 32
Random Seed	Fixed seed 42 across all backbone training and bootstrap runs
Evaluated Datasets	ViHSD test split (n = 6,680), Davidson test split (n = 2,478), Mixed stream (n = 9,158)

9.1 Lexicon Leakage Audit

Table 23 reports the overlap between lexicon entries and held-out train/validation/test splits. All lexicon entries originate from the training split only; no entry is sourced from validation or test.

Table 23. Lexicon leakage audit across train/validation/test splits.

Lexicon	Terms	Bad split rows	Test comment matches	Fired in test not train
VI (`verified_lexicon.csv`)	668	0	13	0
EN (`verified_lexicon_en_combined.csv`)	818	0	0	0

10 Complete List of Supplementary Tables

1. Table 1 — Language routing confusion matrix
2. Table 2 — Language detection accuracy on clean held-out sets (ViNLI + Wortschatz)
3. Table 3 — Sequential latency breakdown for language routing
4. Table 4 — Vietnamese Rule Engine component ablation study
5. Table 5 — Standalone Rule Engine confusion matrix breakdown
6. Table 6 — Per-class metrics for standalone Rule Engine
7. Table 7 — Performance metrics for Vietnamese backbones (M1–M6)
8. Table 8 — Performance metrics for English backbones
9. Table 9 — Multi-class ROC-AUC metrics across backbones
10. Table 10 — Comprehensive decision fusion variants comparison
11. Table 11 — Calibration diagnostics (ECE, Brier score, NLL)
12. Table 12 — Full end-to-end pipeline benchmark results
13. Table 13 — Language-stratified Hybrid E2E performance
14. Table 14 — Full stream class-wise performance metrics

15. Table 15 — Normalized confusion matrix for Hybrid E2E
16. Table 16 — Paired statistical significance tests (Bootstrap & McNemar)
17. Table 17 — Error overlap distribution across backbones
18. Table 18 — Class prior-reweighting oracle diagnostic
19. Table 19 — Performance stratification across surface forms
20. Table 20 — Ground-truth class breakdown of 504 blind-spot samples
21. Table 21 — Summary of decision changes introduced by Hybrid Fusion
22. Table 22 — Hardware, software, and protocol parameters
23. Table 23 — Lexicon leakage audit across train/validation/test splits